

ROBOTICS AND INTELLIGENT SYSTEMS (AIT304)

QUESTION BANK BY MODULE

MODULE I: ROBOT CONFIGURATIONS AND CHARACTERISTICS

Part A Questions (3 Marks)

1. State the three laws of a robot and give its applications. (2023 June, 2024 May) **(3) [REPEATED]**
2. What are the three degrees of freedom associated with the arm and body motion of a robot? (2023 June) **(3)**
3. Briefly describe the Dynamic characteristics of Mobile robot (2024 Dec) **(3)**
4. Categorise the various types of grippers used in robot manipulators (2024 Dec) **(3)**
5. List and explain the features of four basic wheel types used for a robotic application. (2024 Jan) **(3)**
6. List and define the factors influencing the design of a robot. (2024 Jan) **(3)**
7. Explain cylindrical and spherical robot configurations. (2024 May) **(3)**

Part B Questions (14 Marks)

1. (2023 June)
 - a) Explain the four basic robot configurations classified according to the coordinate system. **(10) [REPEATED in 2024 May]**
 - b) Sketch a robot wrist and indicate 3 DOF associated with the wrist. **(4)**
2. (2023 June)
 - a) Briefly explain the dynamic characteristics of robots. **(8)**
 - b) What is the work envelope of a robot? Sketch two views to indicate the work envelope of a Cartesian robot. **(6)**
3. (2024 Dec)
 - a) Explain the general features of wheeled, legged and aerial robots. **(9) [REPEATED in 2024 May]**
 - b) Draw and Explain RPP and RRR robot configuration **(5)**
4. (2024 Jan)
 - a) Explain the anatomy of a robotic manipulator. **(6)**
 - b) Describe the various joint configurations possible in a robotic manipulator. **(8)**
5. (2024 Jan)
 - a) Explain the principle of operation of various grippers used in robotic systems. **(8)**
 - b) Explain the general features of a legged robot and differentiate the terms static and dynamic stability in the context of a legged robot. **(6)**

6. (2024 May)

- a) Explain the four basic robot configurations classified according to the coordinate system. **(8)**
[REPEATED from 2023 June]
- b) Draw and Explain different types of grippers used in mobile robots **(6)**

7. (2024 May, 2024 Dec)

- a) Explain the general features of wheeled, legged and aerial robots. **(9) [REPEATED]**
- b) Assume an object of mass 140 kg is to be lifted up with an acceleration of 10 m/s^2 . Calculate the gripper force required for the operation, if coefficient of friction between contact surfaces is 0.2, number of fingers in gripper is 2 and acceleration due to gravity is 9.8 m/s^2 **(4-5)**
[REPEATED]

MODULE II: ACTUATORS AND SENSORS

Part A Questions (3 Marks)

1. Define the term sensor, and differentiate the same from the transducer. (2023 June, 2024 May) **(3)**
[REPEATED]
2. Compare CCD camera with CMOS Camera. (2023 June, 2024 Dec) **(3) [REPEATED]**
3. Explain Dynamic range, Linearity and Resolution of a sensor/Sensor. (2024 Dec, 2024 Jan, 2024 May) **(3) [REPEATED]**
4. Explain the camera sensor hardware interfacing (2024 Dec) **(3)**
5. Position sensors are not preferred to obtain the velocity and acceleration, Give reason. (2024 Jan) **(3)**

Part B Questions (14 Marks)

1. (2023 June)

- a) A mobile robot is designed for unidirectional motion with constant velocity. Design a mechanism to make the robot move in forward and reverse directions with controllable speed and suggest the sensors and motion control drives required to achieve the same. **(10)**
[REPEATED in 2024 Dec]
- b) Differentiate Stepper Motor and Servomotor concerning the construction and applications. **(4)**
[REPEATED in 2024 May]

2. (2023 June)

- a) Describe the characteristics of sensors. Explain any one sensor used in robotics with the help of neat diagrams. **(7)**
- b) Explain the working of the touch sensor used in robotic applications with a neat sketch. **(7)**

3. (2024 Dec)

- a) Explain the working principle of Hall effect sensor and any one application of the same. **(9)**
[REPEATED in 2024 Jan, 2024 May]

- b) Illustrate the significance of the PID controller with a neat block diagram. **(5)**
4. (2024 Dec)
- a) Explain the working of any two position sensors with neat sketch. **(6)**
 - b) A mobile robot is designed for unidirectional motion with constant velocity. Design a mechanism to make the robot move in forward and reverse directions with controllable speed and suggest the sensors and motion control drives required to achieve the same. **(8) [REPEATED from 2023 June]**
5. (2024 Jan)
- a) Explain the various sensor characteristics to be considered in the choice of the sensor for robotic application. **(6)**
 - b) Explain the working principle of Hall effect sensor and any one application of the same. **(8) [REPEATED from 2024 Dec]**
6. (2024 Jan)
- a) Compare the performance characteristics of a hydraulic and pneumatic actuator. **(6)**
 - b) Explain with a neat circuit diagram the working principle behind a permanent magnet stepper motor **(8)**
7. (2024 May)
- a) Explain the working principle of Hall effect sensor and any one application of the same. **(6) [REPEATED from 2024 Dec, 2024 Jan]**
 - b) Draw and Explain position and velocity internal sensors **(8)**
8. (2024 May)
- a) Briefly describe the speed and direction control of DC motor using H Bridge and PWM **(6)**
 - b) Differentiate Stepper Motor and Servomotor concerning the construction and applications. **(8) [REPEATED from 2023 June]**

MODULE III: ROBOT VISION AND KINEMATICS

Part A Questions (3 Marks)

1. Give the importance of image processing analysis in robotic vision applications. (2023 June) **(3)**
2. Define the terms Steerability, and Maneuverability. (2023 June) **(3)**
3. Differentiate between holonomic and nonholonomic robots. (2024 Dec, 2024 Jan, 2024 May) **(3) [REPEATED]**
4. How will you represent the position and orientation of a wheeled mobile robot? (2024 Dec) **(3)**
5. Write the significance of differential drive, in mobile robotics? (2024 Jan) **(3)**
6. Briefly Describe Sensing, Pre-processing, Segmentation (2024 May) **(3)**

Part B Questions (14 Marks)

1. (2023 June)

- a) Derive the equation for the representation of a pure translation with respect to frame movement in space. **(8)**
- b) Write a short note on the following i. Pre-Processing ii. Segmentation and iii. Recognition. **(6)**

2. (2023 June)

- a) A Frame B is rotated 90° about the z-axis, then translated 2 and 4 units relative to the n and o-axes respectively, then rotated another 90° about the n-axis, and finally, 90° about the y-axis. Find the new location and orientation of the frame. **(8)**
- b) What is an omnidirectional robot? Explain two configurations to set up an omnidirectional robot. **(6) [REPEATED in 2024 Dec, 2024 May]**

3. (2024 Dec)

- a) Outline the seven stages of robot vision **(9) [REPEATED in 2024 Jan, 2024 May]**
- b) A frame F has been moved 10 units along y axis and 5 units along z axis of reference frame. Find new location of Frame **(5)**

4. (2024 Dec)

- a) What is meant by transformation and represent pure translation, pure rotation and combined transformation **(8)**
- b) What is an omnidirectional robot? Explain two configurations to set up an omnidirectional robot. **(6) [REPEATED from 2023 June]**

5. (2024 Jan)

- a) Explain the seven stages of robotic vision. **(10) [REPEATED as "Outline..." in 2024 Dec, 2024 May]**
- b) Differentiate and explain the working of a CCD and CMOS camera in a robotic vision system **(4)**

6. (2024 Jan)

- a) List the different co-ordinate frames used in robotic systems and explain the need for mapping or transformation of co-ordinate systems for the same. **(6)**
- b) Frame {1} and {2} have coincident origins and differ only in orientation. Frame {2} is initially coincident with frame {1}. Certain rotations are carried out about the axis of the fixed frame {1}. First rotation about X axis by 45° then about Y axis by 30° and finally about X axis by 60° . Obtain the equivalent rotation matrix 1R_2 . **(8)**

7. (2024 May)

- a) Outline the seven stages of robot vision. **(14) [REPEATED from 2024 Dec, 2024 Jan]**
- OR
- a) Derive the kinematic model of a differential drive mobile robot. **(7)**

- b) What is an omnidirectional robot? Explain two configurations to set up an omnidirectional robot. **(7) [REPEATED from 2023 June, 2024 Dec]**

MODULE IV: LOCALIZATION AND MAPPING

Part A Questions (3 Marks)

1. What do you mean by Robot Localization? And list out the challenges in localization. (2023 June) **(3)**
2. State the applications and advantages of SLAM robots. (2023 June, 2024 Dec) **(3) [REPEATED]**
3. Identify mobile robot localization problems. (2024 Jan) **(3)**
4. Describe the concept of mobile robot localization with suitable Block diagram. (2024 Jan) **(3)**
5. Identify the 2 mobile robot localization problems. (2024 May) **(3)**
6. How will you represent the error model for odometric position estimation (2024 May) **(3)**

Part B Questions (14 Marks)

1. (2023 June)
 - a) Apply the concept of Accurate Odometry and Error Modelling for a Mobile Robot with a neat sketch. **(7) [REPEATED in 2024 Dec]**
 - b) Discuss in detail about the Map representation and list the current challenges in map representation **(7)**
2. (2023 June)
 - a) How the probabilistic map based localization can be achieved? And suggest the best filtering method for achieving the same. **(10) [REPEATED in 2024 May]**
 - b) List and explain the Decomposition strategies used in map representation. **(4) [REPEATED in 2024 May]**
3. (2024 Dec)
 - a) Describe the concept of mobile robot localization with suitable Block diagrams **(7)**
 - b) Explain the challenges of localization **(7)**
4. (2024 Dec)
 - a) Give the mathematical definition of SLAM and elaborate the concept. **(6) [REPEATED in 2024 Jan]**
 - b) Apply the concept of Accurate Odometry and Error Modelling for a Mobile Robot with a neat sketch. **(8) [REPEATED from 2023 June]**
 - OR
 - a) Give the mathematical definition of SLAM and elaborate the concept. **(6) [REPEATED in 2024 Jan]**
 - b) Compare and Contrast graph-based and particle SLAM **(8)**

5. (2024 Jan)

- a) Define the term odometry and explain the various error sources in odometric data. **(4)**
- b) Derive error model for odometric position estimation. **(10) [REPEATED in 2024 May]**

6. (2024 Jan)

- a) Give the mathematical definition of SLAM and elaborate the concept. **(6) [REPEATED in 2024 Dec]**
- b) Compare and Contrast graph-based and particle SLAM **(8)**

7. (2024 May)

- a) How Kalman Filter method can be used in localization of mobile robots **(7)**
- b) Derive error model for odometric position estimation. **(7) [REPEATED from 2024 Jan]**

8. (2024 May)

- a) How the probabilistic map based localization can be achieved? And suggest the best filtering method for achieving the same. **(10) [REPEATED from 2023 June]**
- b) List and explain the Decomposition strategies used in map representation. **(4) [REPEATED from 2023 June]**
- OR
- b) Why does SLAM work better with wheel odometer data available? **(6)**

MODULE V: PATH PLANNING AND CONTROL

Part A Questions (3 Marks)

1. Differentiate breadth-first search and depth-first search algorithms. (2023 June) **(3)**
2. What do you mean by control decomposition? (2023 June, 2024 Dec, 2024 May) **(3) [REPEATED]**
3. Explain the Bug algorithm for obstacle avoidance (2024 Dec, 2024 Jan) **(3) [REPEATED]**
4. Compare any two traversal algorithms. (2024 Jan, 2024 May) **(3) [REPEATED]**
5. Write short notes on Potential field based path planning (2024 May) **(3)**

Part B Questions (14 Marks)

1. (2023 June)

- a) A neural network can be trained to process sensory data from cameras, Lidar sensor, or other sensors. The network learns to identify objects, obstacles, and landmarks and generates appropriate actions based on this information. Through iterative training and feedback, the neural network can improve its navigation capabilities, adapting to various environments and optimizing its path planning, obstacle avoidance, and decision-making processes of a robot. Summarise the above with suitable sketches. **(10)**
- b) Explain Dijkstra's algorithm with a suitable example. **(4) [REPEATED in 2024 Dec, 2024 Jan, 2024 May]**

2. (2023 June)

- a) Discuss the concept of Modularity for code reuse and sharing. **(8) [REPEATED in 2024 May]**
- b) Give the assumption to be made while applying the Bug algorithm for obstacle avoidance. **(6)**

3. (2024 Dec)

- a) Illustrate the working of Dijkstra's algorithm with a suitable example. **(7) [REPEATED from 2023 June]**
- b) Illustrate the Incorporation of Neural network approach in Robot navigation? List its advantages **(7)**

4. (2024 Dec)

- a) Explain Deterministic graph search algorithms in detail **(7)**
- b) With an example, elaborate the process of horizontal decomposition of a robotic control system. **(7) [REPEATED in 2024 Jan]**

5. (2024 Jan)

- a) Explain the potential field method-based path planning in robotic systems **(6)**
- b) Explain Dijkstra's algorithm with a suitable example. **(8) [REPEATED from 2023 June, 2024 Dec]**

6. (2024 Jan)

- a) Explain the need for control decomposition in a robotic system. **(6)**
- b) With an example, elaborate the process of horizontal decomposition of a robotic control system. **(8) [REPEATED from 2024 Dec]**

7. (2024 May)

- a) Explain Dijkstra's algorithm with a suitable example. **(7) [REPEATED from 2023 June, 2024 Dec, 2024 Jan]**
- b) Compare and contrast local and global Dynamic window approaches in Obstacle avoidance. **(7)**

8. (2024 May)

- a) Discuss the concept of Modularity for code reuse and sharing. **(8) [REPEATED from 2023 June]**
- b) Why does SLAM work better with wheel odometer data available? **(6)**